DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

Autonomous Institute under VTU, Belagavi | Approved by AICTE,New Delhi Recognized by Govt. of Karnataka

NACC Accredited Institution*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belgavi)

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DEPARTMENT OF CSE(DATA SCIENCE)

A Mini Project Report On

"DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS"

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CERTIFICATE

This is to certify that the project work entitled "DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS" is a bonafide work carried out by **{ Ranjita M}** in partial fulfillment for the award of degree of **Bachelor Degree in CSE(Data Science)** in the VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Belagavi during the academic year 2025-2026. It is certified that all corrections and suggestions indicated for internal assessment have been incorporated in the report deposited in the library. The project has been approved as it satisfies the academic requirements in respect of mini project work prescribed for a Bachelor of Engineering Degree.

Signature of project guide	Signature of Coordinator	Signature of HoD
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DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS

ABSTRACT

Migrant workers constitute one of the most mobile and vulnerable segments of the workforce, often experiencing significant barriers in accessing continuous and reliable healthcare services. Their frequent relocation, irregular work schedules, and lack of proper documentation result in fragmented medical histories and delayed treatment, ultimately affecting their overall well-being. Traditional healthcare models fail to support this population effectively, as they rely heavily on physical records, location-based services, and limited communication between healthcare providers. This gap highlights the urgent need for a portable, integrated, and technology-driven solution that ensures consistent health monitoring and timely medical support for migrant communities.

The **Digital Health Management System for Migrant Workers** is developed to address these challenges by offering a centralized digital platform that securely stores and manages personal health records. The system utilizes electronic health records (EHR), cloud-based storage, and mobile accessibility to create a portable health identity for each worker. Through this platform, migrant workers can access their medical history, prescriptions, laboratory reports, immunization details, and ongoing treatment plans from any geographical location. Healthcare providers can retrieve updated patient data instantly, enabling accurate diagnosis and ensuring continuity of care despite frequent mobility.

In addition to record management, the system incorporates features such as automated health reminders, appointment scheduling, real-time alerts, and information on nearby healthcare facilities. These functionalities promote preventive healthcare practices and reduce the likelihood of medical emergencies caused by delayed treatment. The platform also supports multilingual interfaces, making it accessible to workers from diverse linguistic backgrounds.

DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS

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DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS

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DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS

Chapter 1

INTRODUCTION

Migrant workers often face significant challenges in accessing consistent healthcare due to frequent movement across regions. Their medical records are usually scattered, incomplete, or not carried with them, causing delays in treatment. Traditional paper-based systems fail to support continuity of care for this highly mobile population. As a result, health information becomes fragmented and unreliable over time. This affects diagnosis accuracy and increases the burden on healthcare providers. To address these issues, digital solutions are becoming increasingly important. A unified digital platform can greatly improve access and efficiency in migrant healthcare.

The **Digital Health Record Management System for Migrant Workers** aims to centralize and secure health information in a user-friendly digital format. It provides a structured way to store medical history, prescriptions, lab reports, and vaccination details. Through unique IDs or QR codes, migrant workers can access their records from any location. Healthcare professionals can quickly retrieve patient data, enabling timely and accurate decision-making. This reduces repetitive tests and improves overall treatment quality. By bridging communication gaps, the system ensures better coordination between healthcare facilities. Ultimately, it supports a smoother and more reliable healthcare experience.

This project focuses on integrating technology to solve long-standing issues related to migrant workers' health documentation. The system emphasizes data security, reliability, and ease of access for both patients and healthcare providers. It also reduces administrative work by eliminating manual record handling. In addition, the platform promotes awareness among migrant workers about their personal health status. The proposed solution aligns with digital health initiatives aiming for inclusive healthcare. By enabling continuity of care regardless of location, the system enhances medical outcomes.

DIGITAL HEALTH RECORD MANAGEMENT SYSTEM FOR MIGRANT WORKERS

1.1 Overview of "Digital Health Records"

This Digital Health Record Management System helps in:

- Secure digital storage of migrant workers' health records
- Easy access to medical history from any location
- Improved continuity of care during frequent relocations
- Reduced duplication of tests and paperwork
- Faster diagnosis and better treatment decisions
- Reduction of manual work and human errors

Thus, this system plays a key role in creating an efficient, accessible, and inclusive digital healthcare ecosystem for migrant workers.

1.2 Background of "Record Management System"

- Migrant workers move frequently, making it hard to maintain proper health records.
- Their medical information is usually paper-based, incomplete, or lost during travel.
- Manual record-keeping is:
    - Slow
    - Error-prone
    - Hard to access from different locations.

1.2.1 Existing System of "Digital Health Records"

- Health records of migrant workers are mostly maintained manually.

- Medical information is stored on paper in different hospitals and clinics.
- Workers often lose their documents when they move to new locations.
- No centralized digital database exists for migrant workers' health records.
- Doctors cannot easily access past medical history during treatment.
- Repeated tests and delayed diagnosis are common due to lack of data.
- Continuity of care is affected when workers keep changing workplaces or regions.

1.2.2 Objectives

- Developing a secure digital platform to store and manage the health records of migrant workers is a centralized system.
- Enable easy and quick access to medical history from any location, improving continuity of care during frequent relocations.
- Implement a user-friendly interface for both healthcare providers and workers to ensure smooth record retrieval and updating.

1.2.3 Scope Of the Project

- Develop a centralized digital platform to securely store, update, and manage the health records of migrant workers.
- Implement a cloud-based database system to ensure easy access to medical history across different locations and healthcare facilities.
- Provide a user-friendly interface for doctors, healthcare staff, and migrant workers to retrieve or update health information efficiently.

Chapter 2

LITERATURE SURVEY

Digital Health Records and Electronic Medical Systems (EHR/EMR Technologies)

Sales-Contini et al. (2024) demonstrated that pulse frequency, laser power, scanning/marking speed, and spot size have a direct effect on the legibility, geometry, and optical contrast of laser-engraved QR/Data Matrix codes. Their experiments on polymer and composite substrates showed that:

- Higher laser power generally increases engraving depth and mark darkness, but excessive power leads to charring, melting, or dimensional distortion.
- Increased pulse frequency tends to smooth the engraved surface but may reduce contrast when the heat-affected zone expands.
- Marking speed affects thermal accumulation, influencing mark uniformity and line-edge sharpness. Slow speeds improve contrast but risk substrate burning.
- The interaction of parameters shifts depending on material properties, such as polymer crystallinity, filler content, and reflectivity.

Additional works in this domain support similar findings:

- X. Liu et al. (2023) reported that the thermal conductivity and optical absorption behavior of polymer composites strongly affect energy coupling during laser marking, impacting QR readability.
- M. Erden et al. (2022) found that optimal laser settings differ sharply between plastics like ABS, PMMA, and PP due to variation in carbonization thresholds, which determines the resulting contrast.
- Studies on laser-marked Data Matrix codes for aerospace composites highlight the importance of dot roundness, module uniformity, and surface roughness, which influence automated vision-based decoders used in tracking systems.

Overall, research consensus indicates that proper parameter optimization produces measurable improvements in mark depth, width consistency, contrast ratio, and decoding reliability, making laser parameter tuning a critical step for industrial traceability applications.

Enhancement/Repair of Damaged QR Codes Using Deep Learning

J. Yi et al. (2024, *MDPI Electronics*) proposed deep neural network frameworks that detect and restore structural damage in QR and Data Matrix codes. Their system integrates:

- Damage area detection using CNN-based segmentation.
- Iterative restoration that reconstructs missing modules based on QR structure rules and learned visual patterns.
- Error correction enhancement, improving decoding success even when traditional Reed–Solomon EC fails.

Their study showed that deep learning improves decoding rates on QR codes affected by:

- Scratches
- Rust and oxidation
- Surface abrasion
- Partial tearing or missing modules

Additional research in QR code restoration further strengthens this area:

- Y. Chen et al. (2023) introduced a GAN-based model that reconstructs missing QR regions while preserving the geometric structure, achieving high fidelity and readability.
- A. Kumar & B. Singh (2022) developed a hybrid CNN-transformer model capable of restoring low-contrast or blurred QR images, commonly found on aged labels or industrial surfaces.
- S. Hsu et al. (2023) explored super-resolution techniques to enhance low-resolution and distorted QR codes, improving module clarity for machine vision systems.
- Research on defect-tolerant QR readers shows that neural networks outperform traditional thresholding-based decoders, especially under uneven lighting or background noise.

Chapter 3

SYSTEM ANALYSIS

3.1 Existing System (present scenario)

In the existing healthcare system, the medical records of migrant workers are mostly maintained manually in hospitals and clinics. Their health information is stored in paper files, prescription slips, or handwritten notes that are easily lost, damaged, or misplaced during travel. Because migrant workers frequently relocate for employment, hospitals in new locations often have no access to their previous medical history, making diagnosis slow and inaccurate. There is no centralized digital platform where a worker's health records can be accessed from any location, resulting in fragmented and incomplete health data. Manual record-keeping leads to difficulties in retrieving past records, repeating tests, and verifying ongoing treatments. This system is time-consuming, error-prone, and unreliable, especially during medical emergencies. The absence of digital identification or unique health IDs makes tracking past treatments, allergies, and ongoing medications very difficult. Overall, the existing healthcare record system does not support portability, real-time access, or proper continuity of care for migrant workers.

3.2 Limitations of the Existing System

- Existing system is manual and paper-based, making it slow and unorganized.
- High chances of human error in maintaining and retrieving records.
- No centralized database to access migrant workers' health history.
- Medical records can be lost, damaged, or misplaced during travel.
- No digital identification is available for individual migrant workers.
- Past medical history and treatment records are difficult to track.
- Doctors must repeat diagnostic tests due to lack of previous data.
- Emergency care becomes difficult without quick access to patient history.
- System does not support remote access or digital continuity of care.

3.3 Feasibility Study

1. Technical Feasibility

- Digital record systems and cloud storage are easily available.
- Secure login and encryption ensure safe data handling.
- Unique IDs or QR codes can be used for easy access.
- Compatible with existing hospital workflows.

2. Economic Feasibility

- Development cost is moderate.
- Reduces long-term costs of paperwork and repeated tests.
- Minimizes financial losses caused by missing data.
- Cost-effective over time.

3. Operational Feasibility

- Easy to learn and use for hospital staff.
- Reduces manual workload.
- Improves speed and accuracy of treatment.
- Supports access from any location.

4. Schedule Feasibility

- Can be developed within a short timeline.

- Implementation possible in phases.
- Easy to upgrade later.

5. Social Feasibility

- Improves healthcare access for migrant workers.
- Reduces treatment errors.
- Supports digital healthcare initiatives.

3.4 System Requirements

1. Hardware Requirements

- Computer / Laptop
- Smartphone or tablet (for worker access)
- Cloud server or storage space

2. Software Requirements

- Operating System: Windows / Linux
- Programming: Python / Java / JavaScript
- Database: MySQL / MongoDB
- Framework: Django / Flask / Node.js
- Browser: Chrome / Edge

Chapter 4

SYSTEM DESIGN

4.1 General

The Digital Health Record Management System for Migrant Workers is designed to provide secure, portable, and accessible medical records for mobile populations. Each worker receives a unique digital identity (ID/QR) linked to their encrypted electronic health record (EHR). Mobile and clinic applications allow authorized health professionals and workers to view, update, and upload medical data such as diagnoses, prescriptions, lab reports, and vaccination records. Data captured at the edge (mobile devices or clinic terminals) is synchronized with a centralized cloud backend that stores patient metadata, health history, and audit logs. Interoperability standards (FHIR/HL7) enable data exchange with hospitals, labs, and government health systems. Analytics and rule-based engines monitor records for critical alerts (drug interactions, overdue vaccinations), assist with continuity of care, and support reporting. The system improves record portability, reduces redundant tests, speeds clinical decisions, and strengthens health coverage for migrant workers.

4.2 Architectural Overview

The architecture is organized into four main layers:

1. **Digital ID & Portable Records:** Unique IDs/QRs or biometric link for each migrant worker to enable quick look up and portability.
2. **Edge Access Units:** Mobile/clinic apps and kiosks for data capture, offline caching, and immediate access during consultations.
3. **Secure Data Transmission:** Encrypted communication (TLS/HTTPS, VPN) for syncing edge devices with central services.

4. **Central Backend & Dashboards:** Cloud-hosted EHR database, analytics engine, reporting dashboard, and API gateway for external integration.

4.3 Design Components & Their Role

- **ID Issuance Module** — generates unique health IDs, QR codes, and optional biometric enrollment for reliable identity linkage.
- **Patient Mobile App / Portal** — allows workers to view records, share QR/ID, and upload documents or images.
- **Clinic / Provider App** — enables registration, record updates, prescriptions, and scanning of worker IDs.
- **Data Capture & Validation** — forms, attachments, and metadata validation to ensure structured and accurate entries.
- **Offline Cache & Sync** — local storage on devices that queues updates and securely syncs when connectivity returns.
- **API & Interoperability Layer** — RESTful/FHIR APIs for data exchange with hospitals, labs, insurance, and government systems.
- **Central EHR Database & Object Store** — relational/NoSQL store for structured records and object storage for reports/images.
- **Authentication & Access Control** — role-based access, multi-factor authentication, and audit logging to protect PHI.
- **Encryption & Key Management** — data-at-rest and in-transit encryption with secure key management.
- **Analytics & Alert Engine** — rule-based and ML-assisted modules for medication alerts, chronic-care follow-up, and population reports.
- **Admin Dashboard & Reporting** — real-time dashboards for health workers, program managers, and public-health officials.
- **MLOps & Data Pipeline** — dataset management, model training, validation, and safe deployment processes for predictive features.

4.4 System Workflow

- **ID Enrollment**: Migrant worker is registered and issued a unique digital ID/QR and optional biometric token.
- **Data Capture**: At clinics or via mobile app, clinicians record visits, diagnoses, prescriptions, and upload lab reports.
- **Edge Validation & Cache**: Device validates entries locally and stores them if offline.
- **Secure Sync**: When online, the edge unit encrypts and transmits records to the central backend.
- **Central Storage & Processing**: Backend stores records, runs analytics, updates longitudinal history, and maintains audit trails.
- **Alerts & Follow-up**: Alerts (allergy warnings, medication conflicts, overdue screenings) are generated and notifications sent to patients/providers.
- **Reporting & Integration**: Health managers view dashboards and export reports; external systems access data via secure FHIR APIs.

4.5 Technology Stack Design

- **Backend:** Python (Django / Flask) or Java (Spring Boot) for APIs and business logic.
- **Database:** PostgreSQL / MySQL for structured EHR, object storage (S3-compatible) for files.
- **Frontend / Mobile:** React / React Native or Flutter for responsive web and mobile apps.
- **Interoperability:** HL7 FHIR-compliant API layer.
- **Security:** OAuth2 / JWT, TLS, AES encryption for data protection.
- **Cloud / Hosting:** AWS / Azure / GCP or private cloud for scalability and availability.

- **Analytics / ML:** Python (Pandas, scikit-learn, TensorFlow) for alerts and predictive modules.

This stack balances reliability, interoperability, and ease of maintenance.

4.6 Design Advantages

- **Portable Records:** Enables continuous care irrespective of geographic moves.
- **Faster Decisions:** Immediate access to past history reduces diagnostic delays.
- **Reduced Duplication:** Fewer repeated tests and lower healthcare costs.
- **Improved Safety:** Alerts prevent medication conflicts and missed vaccinations.
- **Scalable & Interoperable:** Integrates with existing health systems and scales to large populations.
- **Secure & Auditable:** Strong access controls, encryption, and audit logs protect patient privacy.

The proposed design creates a resilient, secure, and user-friendly system that improves healthcare continuity, reduces administrative burden, and enhances outcomes for migrant workers.

Chapter 5

IMPLEMENTATION

5.1 Implementation Details

The implementation begins by registering migrant workers and generating unique digital IDs or QR codes linked to their personal health profiles. Healthcare providers use mobile or web applications to record medical visits, diagnoses, prescriptions, and lab reports directly into the system. When a consultation occurs, the worker's ID/QR is scanned to instantly retrieve their medical history. Edge modules on mobile devices validate and temporarily store the data, ensuring functionality even in low-connectivity regions. The collected information is then securely synchronized with the central cloud server through encrypted transmission channels. The backend processes and stores the updated health records, triggers alerts for critical conditions, and integrates with external health services or government portals for additional care coordination.

5.2 Digital ID Generation and Patient Enrollment

Each migrant worker is enrolled into the system with a unique digital identifier or QR code. This ID is permanently linked to their health record to ensure accurate identity mapping. During registration, basic demographic data, medical history, allergies, and vaccination records are captured. The QR code is tested for readability across devices to ensure consistent access during treatment.

5.3 Health Data Capture and Edge Processing

Healthcare staff use mobile applications or clinic terminals to capture medical data during consultations. The worker's QR code is scanned to retrieve their record, and new information such as symptoms, diagnosis, prescriptions, or lab uploads is entered. Edge processing components validate the data, perform local encryption, and temporarily store it when offline. Once connectivity is restored, the cached data is synchronized automatically with the central server.

5.4 Central Server Integration and Healthcare Automation

The synchronized data—including patient ID, visit details, prescriptions, and medical documents—is transferred securely to the central backend. The server updates the patient's digital health record, performs analytics for alerts or follow-up reminders, and logs all changes for auditing. The backend system also integrates with hospital systems, laboratories, and digital health platforms to support referrals, test reporting, and public health monitoring. Automated notifications are generated for vaccinations, chronic disease follow-ups, medication conflicts, and overdue health screenings

Chapter 6

TESTING

Testing is a critical phase of this project because the system handles sensitive medical information and supports healthcare decision-making for migrant workers. The Digital Health Record Management System must accurately store, retrieve, and update patient records, ensure secure access control, maintain reliable data synchronization, and protect confidentiality at all times. Any error in data handling, identity matching, or record retrieval can lead to incorrect diagnosis and treatment risks. Therefore, the testing process includes unit testing, integration testing, system-level evaluation, and performance validation to ensure accuracy, reliability, security, and robustness under real-world healthcare conditions..

6.1 Unit Testing

Unit testing was performed on individual software components before full system integration. The user registration module was tested using various data inputs to verify successful ID or QR code generation. The login and authentication module was tested to ensure proper role-based access and prevent unauthorized entry. The medical data entry functions, including diagnosis, prescription upload, and record updates, were tested for form validation and data accuracy. The encryption and decryption processes were tested to confirm that sensitive health information remained secured during storage and transmission. Unit testing ensured that each module performed correctly in isolation.

6.2 Integration Testing

After validating individual modules, integration testing was carried out to ensure all components worked together smoothly. The QR/ID scanning module was integrated with the patient record

retrieval system to confirm instant and correct record access. The data entry module was tested with the cloud synchronization service to verify that updated medical records were successfully transmitted and stored.

6.3 System Testing

System testing evaluated the complete solution under realistic healthcare usage scenarios. Simulated patient check-ups were conducted to verify accurate record retrieval, smooth data entry, and proper updating of medical histories. Various conditions were introduced, such as missing network connection, incomplete patient data, and simultaneous access by multiple users, to test system resilience. Offline mode was tested to ensure data caching and later synchronization without loss or corruption. Alerts for high-risk conditions, vaccination reminders, and medication conflicts were evaluated to ensure timely and correct notifications. Dashboard reporting was tested to confirm real-time system updates.

6.4 Performance Testing

Performance testing assessed the speed, scalability, and responsiveness of the system. The system was tested under high user load to evaluate server performance during peak usage. Medical records were retrieved and displayed within seconds, ensuring fast access during patient consultations. The system successfully handled multiple simultaneous data uploads without performance degradation. Network latency tests confirmed stability during weak or fluctuating connectivity conditions. The synchronization engine was verified to efficiently process queued offline data once the network was restored. Overall, the system maintained high performance under diverse operational conditions.

6.5 Validation Results

Validation testing confirmed that the system met its functional and non-functional requirements. Digital IDs and QR codes correctly mapped to patient records, and all medical data—diagnoses, prescriptions, reports—were accurately stored and retrieved. Data security features such as encryption, access control, and audit logs worked reliably. Alerts, reminders, and follow-up notifications triggered correctly, improving continuity of care. The system demonstrated strong

performance in portability, efficiency, and usability. Overall, the testing validated that the Digital Health Record Management System enhances healthcare access, reduces manual errors, improves data reliability, and supports better medical decision-making for migrant workers.

Chapter 7

Result

Login page

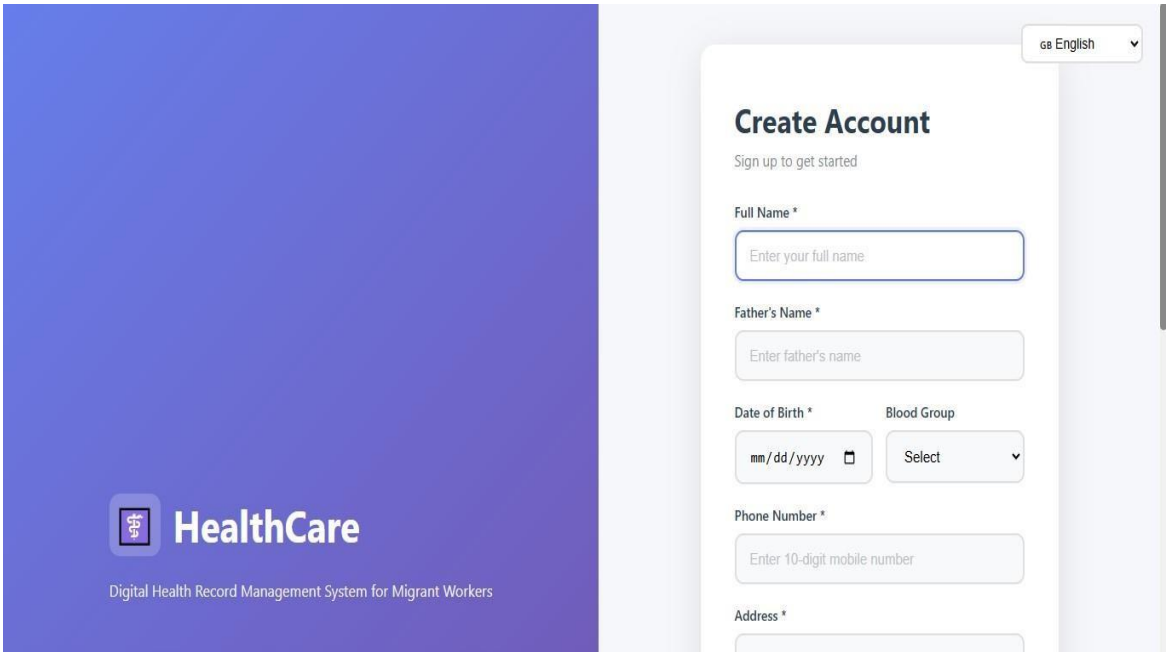

Fig 7.1 Login Page

- The screen shows a Create Account page of the *Digital Health Record Management System for Migrant Workers*.
- The left side displays the project branding with the title **"**HealthCare**"** and a tagline indicating digital health record management.
- The right side contains a **registration form** where users can enter personal details such as full name, father's name, date of birth, blood

group, phone number, and address.

Worker Dashboard:

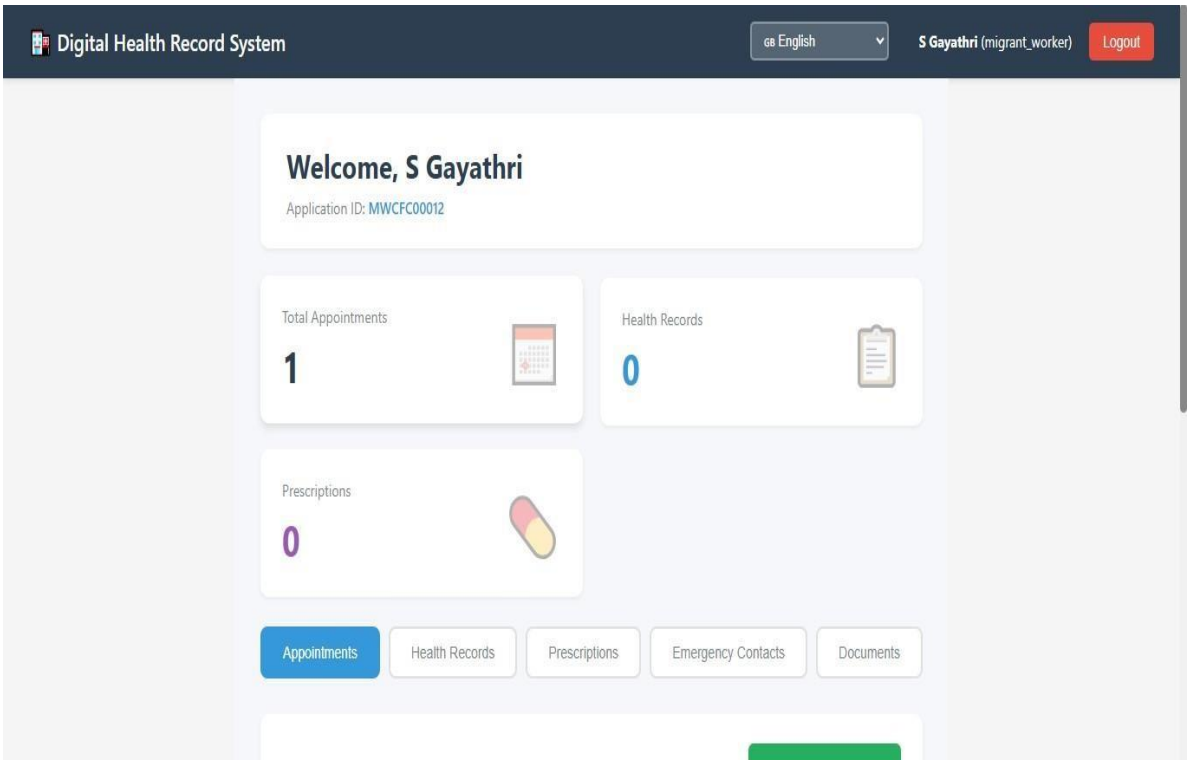

Fig 7.2 Migrant Worker Dashboard

- The system successfully displays a personalized dashboard with the worker's name and unique ID.
- Key health information such as appointments, records, and prescriptions is shown clearly.
- Users can easily navigate through sections like Appointments, Health Records, and Documents.
- The interface is simple, user-friendly, and supports multiple languages

Doctor's Dashboard:

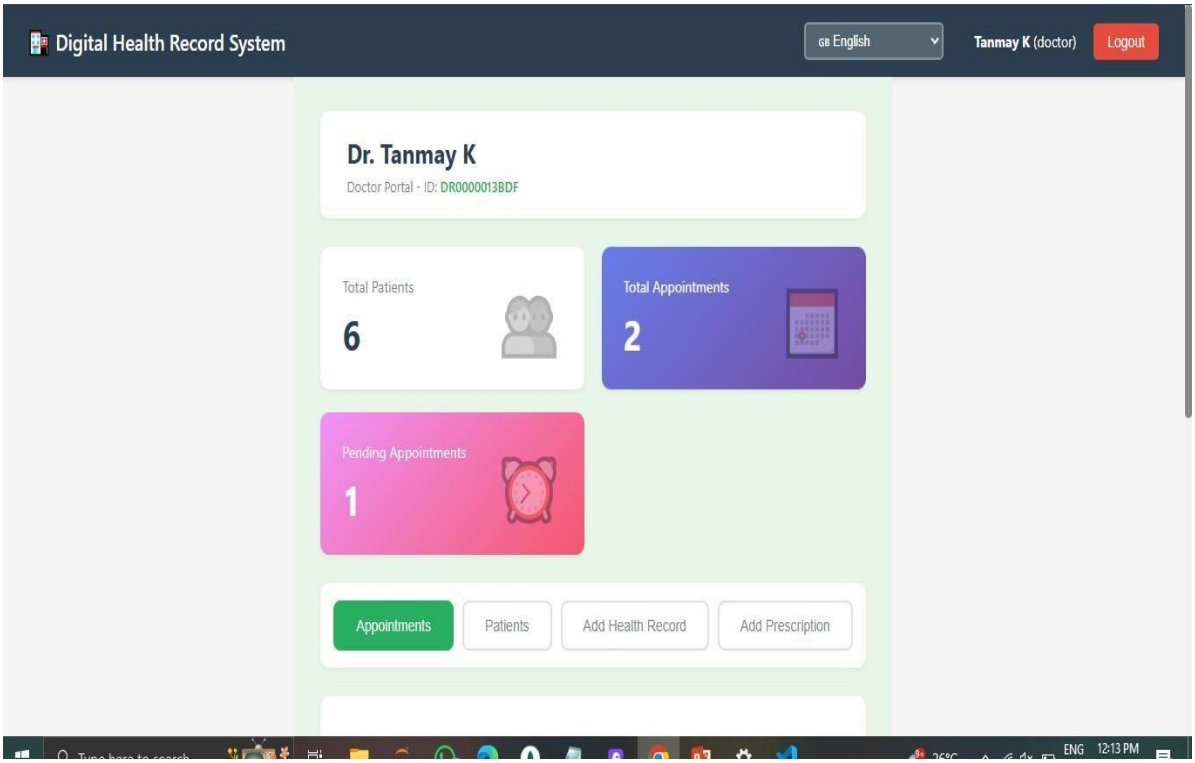

Fig 7.3 Doctor's Dashboard

- screen displays the Doctor Dashboard of the Digital Health Record System.
- The top bar shows language selection, doctor's name and a Logout button for security.
- The dashboard shows the doctor's details, including their Doctor Portal ID.
- Summary cards display important stats: Total Patients (6), Total Appointments (2), and The Pending Appointments (1).

Health Worker Dashboard:

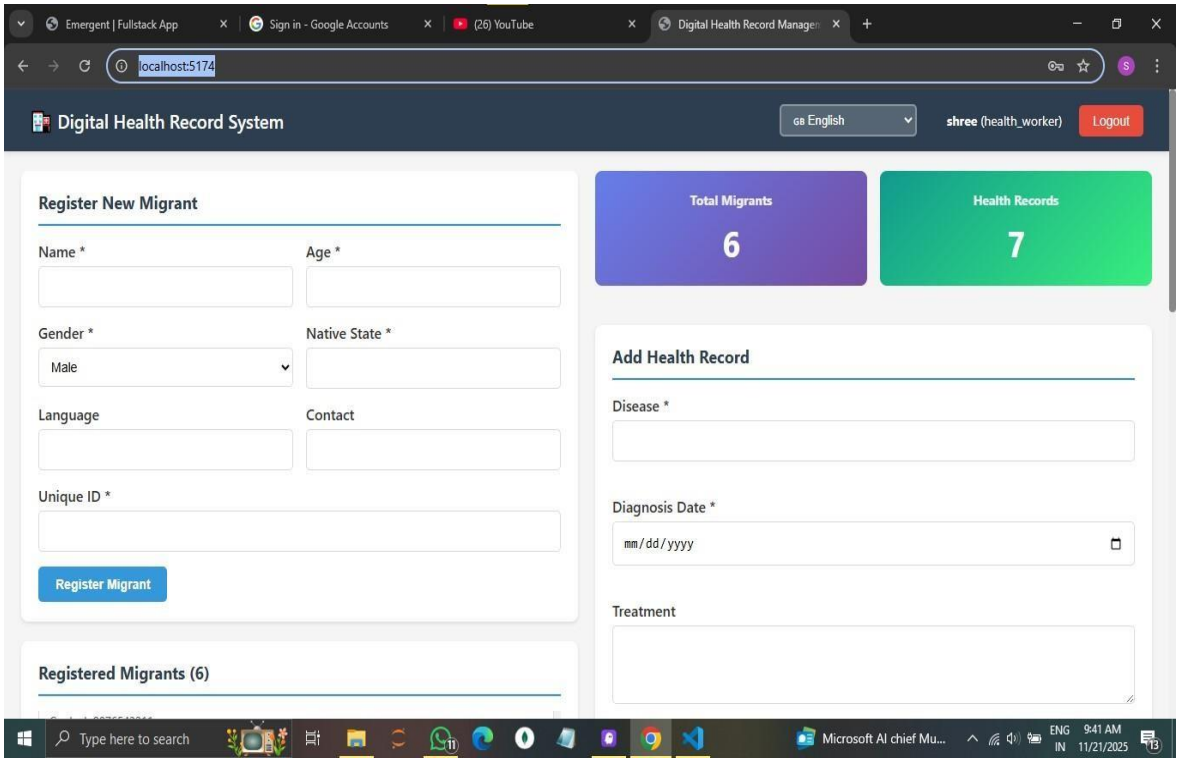

Fig 7.4 Health Worker Dashboard

- The screen shows the **Health Worker Dashboard** of the Digital Health Record System.
- At the top, there is a language selector, the logged-in user's name and a Logout button.
- On the left side, there is a Register New Migrant form with fields such as Name, Age, Gender, Native State, Language, Contact, and Unique ID.
- The Register Migrant button allows the health worker to add new migrant

profiles into the system.

CONCLUSION

The Digital Health Record Management System for Migrant Workers provides a comprehensive and efficient solution to one of the most persistent challenges faced by mobile populations—lack of accessible, portable, and accurate medical records. By digitizing health data and linking it to a unique identity system, the project ensures continuity of care regardless of geographical movement. The system enhances healthcare delivery by enabling quick retrieval of patient history, reducing repeated diagnostic tests, and improving the accuracy of clinical decisions.

Securing cloud-based storage, role-based authentication, and encrypted communication ensure that sensitive medical information is protected at all stages. The integration of mobile and web interfaces makes the system highly accessible to both healthcare providers and migrant workers, even in remote or resource-limited settings. Automated alerts, reminders, and data analytics further support preventive care and timely interventions.

Overall, the project demonstrates how digital technology can bridge the healthcare gap for underserved populations and strengthen public health infrastructure. By improving record management, reducing manual errors, and supporting informed medical decisions, this system contributes significantly to creating an inclusive, reliable, and future-ready digital healthcare ecosystem for migrant workers.

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